Electromagnetic Fields 2020 - Solutions

1a) Ampere's law is $\int_L \mathbf{H} \cdot d\underline{L} = \iint_A [\mathbf{J} + \partial\mathbf{D}/\partial t] \cdot d\underline{a}$. Here
$\mathbf{J}$ is the normal current density and describes the flow of electrons
$\mathbf{J}_D = \partial\mathbf{D}/\partial t$ is the displacement current density and describes the effect of time-varying fields

[2]

Now the constitutive equations give
$\mathbf{J} = \sigma\mathbf{E}$, where σ is the conductivity and $\mathbf{E}$ is the electric field
$\mathbf{D} = \varepsilon\mathbf{E}$, where ε is the conductivity
To compare effects seen with harmonic electric fields we can write
$\mathbf{E} = \underline{E} \exp(j\omega t)$, where ω is the angular frequency
Hence $|\mathbf{J}_D/\mathbf{J}| = \omega\varepsilon/\sigma = \omega\varepsilon_0\varepsilon_r/\sigma$

[2]

Assuming that the dielectric constant of free space is 8.85×10^{-12} F/m, and for intrinsic silicon:
$\sigma = 1.56 \times 10^{-3}$ S/m
$\varepsilon_r = 11.7$, so $\varepsilon = 11.7 \times 8.85 \times 10^{-12} = 1.035 \times 10^{-10}$ F/m
At 1 MHz, we get:
$|\mathbf{J_D}/\mathbf{J}| = 2\pi \times 10^6 \times 1.035 \times 10^{-10} / 1.56 \times 10^{-3} = 0.4169$
At 1 GHz, we get:
$|\mathbf{J}_D/\mathbf{J}| = 2\pi \times 10^9 \times 1.035 \times 10^{-10} / 1.56 \times 10^{-3} = 41.69$

[4]

b) Coaxial cable

Electric field: Gauss' law states that $\iint_A \underline{\mathbf{D}} \cdot d\underline{a} = \iiint_V \rho \, dv$

If we construct a cylindrical Gaussian surface of unit length into the plane of the figure at any radius r greater than the cable radius (see A below), we will enclose equal and opposite charge ±q. In this case there is no net charge enclosed so the RHS must be zero. Consequently, the LHS = 0. Symmetry implies that $\underline{\mathbf{D}}$ is radial and there is no variation with angle. The LHS must then evaluate as $2\pi r D_r$ and hence $D_r = 0$.

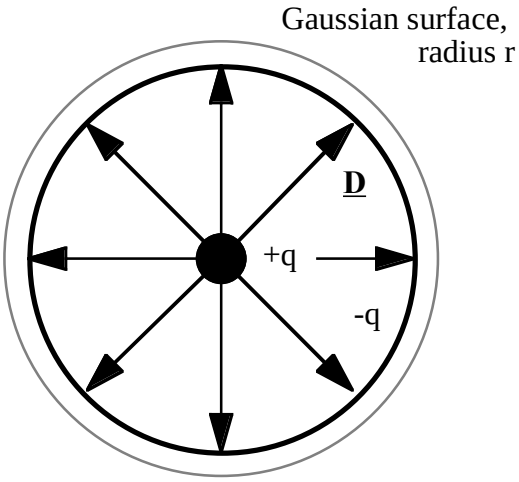
[4]

Magnetic field: Ampere's law states that $\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$

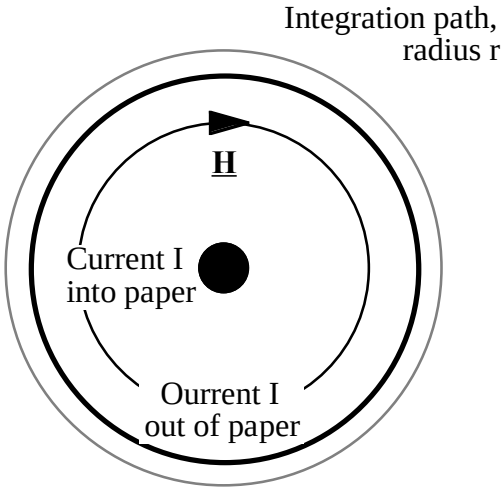
Ignoring the displacement current (since $\underline{\mathbf{D}} = 0$ outside the cable), $\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A \underline{\mathbf{J}} \cdot d\underline{a}$

If we construct a circular integration path at any radius r greater than the cable radius (see B below), we will enclose equal and opposite current ±I. In this case there is no net current enclosed and the RHS must be zero. Consequently, the LHS = 0. Symmetry implies that $\underline{\mathbf{H}}$ is tangential and there is no variation with angle. The LHS must then evaluate as $2\pi r H_\theta$ and hence $H_\theta = 0$.

[4]

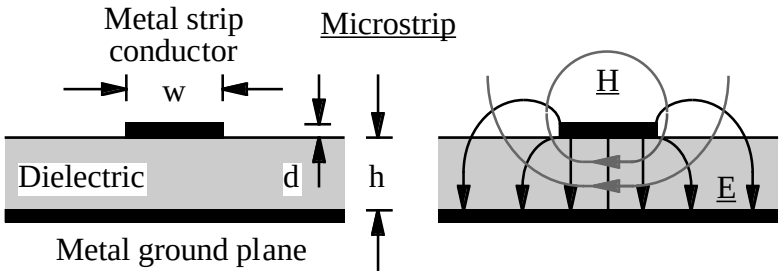


A



B

c) Microstrip layout (bookwork) is as follows:



For microstrip, the electric field between the strip conductor and ground gives rise to a capacitance. The simplest approximation gives a pul capacitance of $C_p \approx \varepsilon w/h$. Including fringing fields (estimated to spread h on either side), $C_p \approx \varepsilon(w + 2h)/h$.

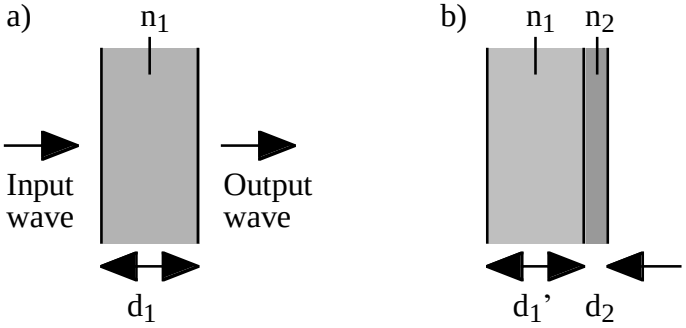
The magnetic field between the conductors gives rise to an inductance. Using Ampere's law, ignoring the distant parts of the field and assuming the remainder is constant, a line integral round a loop gives $Hw \approx I$. The flux density is therefore $B = \mu_0 I/w$. The pul flux linked between the conductors is then $\Phi_p = Bh = \mu_0 Ih/w$ so the pul inductance is $L_p = \Phi_p/I = \mu_0 h/w$.

[4]

The characteristic impedance is $Z_0 = (L_p/C_p)^{1/2} \approx \{\mu_0 h/\varepsilon w(2 + w/h)\}^{1/2}$
Re-arranging, we get $w(2 + w/h) \approx \mu_0 h/\varepsilon Z_0^2$
Hence w is the solution to the quadratic $w^2 + 2hw - \mu_0 h^2/\varepsilon Z_0^2 = 0$
The strip width is then $w = h\{-1 \pm \sqrt{(1 + \mu_0/\varepsilon Z_0^2)}\}$
If $Z_0 = 50\ \Omega$, $h = 1.6$ mm and $\varepsilon_r = 4.4$, the positive solution is $w = 4.4$ mm

[4]

d) Optical transfer functions of slabs and multilayers



Ignoring reflections, the effect of a single transparent slab of thickness d_1 and refractive index n_1 as shown in a) is just a phase delay, so that $E_{out} = E_{in} \exp(-jk_0 n_1 d_1)$ The transfer function of a slab is therefore $\tau_S = \exp(-jk_0\delta_1)$, where $\delta_1 = n_1 d_1$ is the optical thickness.

For a pair of slabs, with the first having refractive index n_1 and thickness d_1' and the second having refractive index n_2 and thickness d_2 as shown in b), the output can be found by multiplication of transfer functions, as:
$E_{out} = E_{in}\, \tau_1 \times \tau_2$, or $E_{out} = E_{in} \exp\{-jk_0(n_1 d_1' + n_2 d_2)\}$
So the overall transfer function is $\tau = \exp(-jk_0\delta)$ with $\delta = n_1 d_1' + n_2 d_2$

[4]

If the transfer function is to remain unaltered, we must have $n_1 d_1' + n_2 d_2 = n_1 d_1$
Hence $d_1' = d_1 - (n_2/n_1)\, d_2$
If $n_1 = 1.5$, $n_2 = 1.33$, $d_1 = 100\ \mu m$ and $d_2 = 10\ \mu m$ we then require:
$d_1' = 100 - (1.33/1.5) \times 10\ \mu m = 91.13\ \mu m$

[4]

e) Optical reflection and refraction

The TE reflection coefficient for normal incidence from a first dielectric with refractive index n_1 onto a second dielectric with refractive index n_2 is $\Gamma_E = (n_1 - n_2) / (n_1 + n_2)$. If $n_1 = 1$:

$\Gamma_E = (1 - n_2) / (1 + n_2)$ so

$(1 + n_2)\,\Gamma_E = (1 - n_2)$ and

$n_2 = (1 - \Gamma_E)/(1 + \Gamma_E)$

Hence if $\Gamma_E = -0.2$, $n = (1 + 0.2) / (1 - 0.2) = 1.2/0.8 = 1.5$

[2]

For incidence at an angle θ_1, the angle of refraction θ_2 can be found using Snell's law:

$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$

Hence, $\theta_2 = \sin^{-1}\{(n_1/n_2) \sin(\theta_1)\}$

If $\theta_1 = 30^o$, $\theta_2 = \sin^{-1}\{(1/1.5) \sin(30)\} = \sin^{-1}(1/3) = 19.47^o$

[3]

Under the same conditions, the TE reflection coefficient is given by:

$\Gamma_E = \{n_1 \cos(\theta_1) - n_2 \cos(\theta_2)\} / \{n_1 \cos(\theta_1) + n_2 \cos(\theta_2)\}$

Hence $\Gamma_E = \{\cos(30) - 1.5 \cos(19.47)\} / \{\cos(30) + 1.5 \cos(19.47)\} = -0.24$

[3]

2. For a particular medium, the variation between the angular frequency ω and the propagation constant k for electromagnetic waves takes the form $\omega^2 = \omega_P^2 + c^2k^2$, where, ω_P is the plasma frequency and c is the velocity of light.

a) The dispersion diagram can be constructed from the following arguments. At angular frequencies above ω_P, real solutions can be found for k, so propagating waves exist in this regime. Below ω_P, no real solutions can be found for k, so propagating waves do not exist.

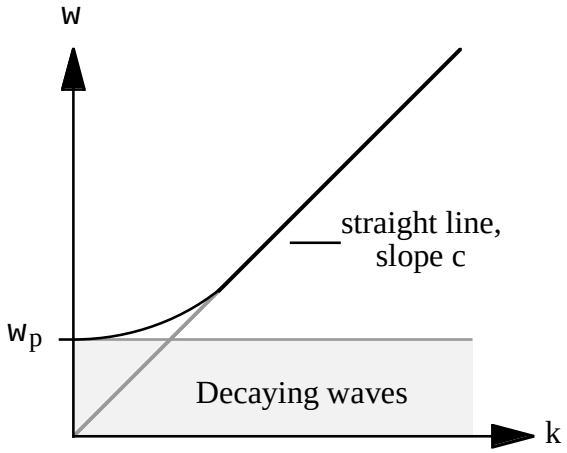
[2]

When k is small, the approximation $\omega^2 = \omega_P^2$ can be made. In this regime, $\omega = \omega_P$, and the dispersion characteristic is therefore a straight horizontal line.

When k is large, the approximation $\omega^2 = c^2k^2$ can be made. In this regime, $\omega = ck$, and the dispersion characteristic is therefore a straight line with slope c.

[2]

The overall dispersion characteristic is a smooth curve linking the low- and high-frequency asymptotes as shown below, with the propagation regimes as indicated.



[2]

b) The phase velocity can be found as $v_{ph} = \omega/k$.
From the dispersion relation, $k = (\omega^2 - \omega_P^2)^{1/2}/c$
Hence $v_{ph} = c\omega/(\omega^2 - \omega_p^2)^{1/2}$

[2]

The group velocity can be found as $v_g = d\omega/dk$
Differentiating the dispersion relation, $v_g = c^2 \cdot 2k \cdot 1/2 \cdot \{(\omega_P^2 + c^2k^2)\}^{-1/2} = c^2k/\omega$
From the dispersion relation, $k = (\omega^2 - \omega_P^2)^{1/2}/c$
Hence, $v_g = c(\omega^2 - \omega_P^2)^{1/2}/\omega$

[2]

When ω is just above ω_P, so that $\omega = \omega_P + \Delta\omega$, we can make the approximations:

$\omega^2 = \omega_P^2 + 2\omega_P\Delta\omega$ and $\omega^2 - \omega_p^2 = 2\omega_P\Delta\omega$. In this regime:

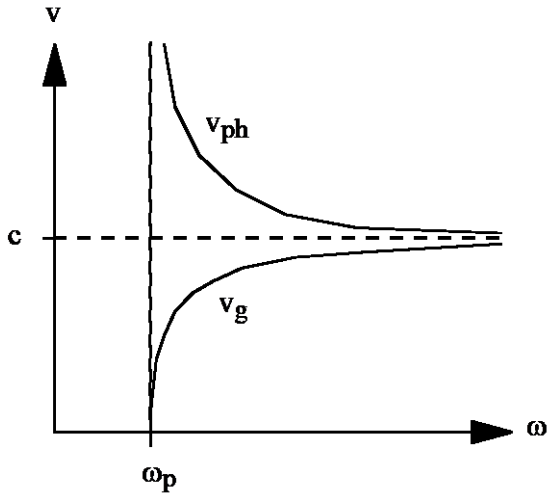
$v_{ph} = c(\omega_P^2/2\omega_P\Delta\omega)^{1/2} = c(\omega_P/2\Delta\omega)^{1/2}$, so the variation of v_{ph} with $\Delta\omega$ is inverse

$v_g = c\{1 - \omega_P^2/(\omega_P^2 + 2\omega_p\Delta\omega)^{1/2} = c(2\Delta\omega/\omega_P)^{1/2}$, so the variation of v_g with $\Delta\omega$ is square root

However, when ω is much greater than ω_P, the approximations $v_{ph} = v_g = c$ can be used

[2]

These results can be plotted as follows:



[2]

c) Rearranging the dispersion equation, we get $k = (\omega^2 - \omega_P^2)^{1/2}/c$, so k is imaginary for $\omega < \omega_P$ and can be written as $k = -j(\omega_P^2 - \omega^2)^{1/2}/c$

[2]

In general, the variation of a wave amplitude with distance z is $A(z) = A_0 \exp(-jkz)$. If k is imaginary, so that k = -jk'', this variation reduces to $A(z) = A_0 \exp(-k''z)$. The amplitude decays exponentially, and the distance in which it falls to 1/e of its initial value is therefore $z = 1/k''$

[2]

If $f_P = 1$ GHz and f = 0.999 GHz, $k = 2\pi \times 10^9 (0.999^2 - 1)^{1/2}/(3 \times 10^8) = \pm j\, 0.2961\ m^{-1}$

The 1/e distance is then $z = 1/0.2961 = 3.376$ m

[2]

3a. The scalar wave equation for electric fields that only vary with radius r is:

$d^2E/dr^2 + (2/r)\, dE/dr + \omega^2\mu_0\varepsilon_0 E = 0$

Differentiating the diverging spherical wave solution $E = (E_0/r) \exp(-jk_0r)$, we get:

$dE/dr = -(E_0/r^2) \exp(-jk_0r) - jk_0(E_0/r) \exp(-jk_0r)$

$d^2E/dr^2 = 2(E_0/r^3) \exp(-jk_0r) + jk_0(E_0/r^2) \exp(-jk_0r) + jk_0(E_0/r^2) \exp(-jk_0r) - k_0^2(E_0/r) \exp(-jk_0r)$ or

$d^2E/dr^2 = 2(E_0/r^3) \exp(-jk_0r) + 2jk_0(E_0/r^2) \exp(-jk_0r) - k_0^2(E_0/r) \exp(-jk_0r)$

Substituting into the wave equation, we get:

$2(E_0/r^3) \exp(-jk_0r) + 2jk_0(E_0/r^2) \exp(-jk_0r) - k_0^2(E_0/r) \exp(-jk_0r)$

$- 2(E_0/r^3) \exp(-jk_0r) - 2jk_0(E_0/r^2) \exp(-jk_0r) + \omega^2\mu_0\varepsilon_0(E_0/r) \exp(-jk_0r) = 0$

Cancelling terms, we get:

$-k_0^2(E_0/r) \exp(-jk_0r) + \omega^2\mu_0\varepsilon_0(E_0/r) \exp(-jk_0r) = 0$

Removing common factors, we then get

$-k_0^2 + \omega^2\mu_0\varepsilon_0 = 0$

[2]

The solution therefore satisfies the wave equation, with propagation constant of $k_0 = \omega^2(\mu_0\varepsilon_0)^{1/2}$

[1]

In a diverging spherical wave, power is carried radially outward, and the irradiance or time-averaged power density is $S_r = |E^2|/2Z_0$, where Z_0 is the characteristic impedance of free space.

Using the spherical wave solution, we get $S_r = E_0^2/2r^2Z_0$

The power passing through a spherical surface of radius r is $P = S_r \,.\, 4\pi r^2 = 4\pi E_0^2/2Z_0$

This expression is independent of r, implying that the wave solution conserves power

[2]

b) Antenna directivity is a measure of the enhancement in power density in the preferred direction compared with the average. Assuming a normalised radiation pattern $F(\theta, \phi)$ the average power density is $1/4\pi \iint_{4\pi} F(\theta, \phi) \sin(\theta)\, d\theta\, d\phi$. Since the maximum in F is unity, the directivity is $D = 1/ \{1/4\pi \iint_{4\pi} F(\theta, \phi) \sin(\theta)\, d\theta\, d\phi\}$

[1]

i) At radius r from a transmitter of power P, the isotropic power density is $S_{iso} = P/4\pi r^2$

If $P = 2 \times 10^3$ W and $r = 10^4$ m, $S_{iso} = 2 \times 10^3/(4\pi \times 10^8)$ W/m^2 = 1.592 μW/m^2

[2]

ii) Assuming an antenna efficiency η and directivity D, the real power density is $S_{real} = \eta D\, S_{iso}$.

If $\eta = 0.8$ and $D = 100$, we obtain $S_{real} = 0.8 \times 100 \times 1.592$ μW/m^2 = 0.1273 mW/m^2

[2]

c) To derive an expression for a paraxial wave, start with the solution $E = (E_0/r) \exp(-jk_0r)$

Assuming that $r = \sqrt{(z^2 + R^2)}$ and $R << r$, we may write:

$1/r \approx 1/z$

$r = z\sqrt{(1 + R^2/z^2)} \approx z(1 + R^2/2z^2) \approx z + R^2/2z$

Hence $E \approx (E_0/z) \exp\{-jk_0(z + R^2/2z)\} \approx E_0'(z) \exp(-jk_0R^2/2z)$

[3]

The phase of the wave varies parabolically with the transverse radial distance R.

[1]

For a wave converging on a point z away, the expression modifies to $E \approx E_0'(z) \exp(+jk_0R^2/2z)$

[1]

d) If a paraxial wave is incident on a lens from an object point u away, the input field is:

$E_{in} = E_0' \exp(-jk_0R^2/2u)$

If the lens transfer function is $\tau_L = \tau_S \exp(jk_0R^2/2f)$, where f is the focal length, the output field is:

$E_{out} = E_{in}\tau_L = E_0'\tau_S \exp(-jk_0R^2/2u) \exp(jk_0R^2/2f)$, or

$E_{out} = E_0'\tau_S \exp(+jk_0R^2/2v)$ where $1/v = 1/f - 1/u$

A lens therefore transforms one paraxial wave into another, and the object and image distances are related by the imaging equation $1/u + 1/v = 1/f$.

[3]

If u = 200 mm and f = 100 mm, $1/v = 1/100 – 1/200 = 1/200$; hence v = 200 mm.

[1]

In general, the magnification is v/u. In this case, it is clearly unity.

[1]